The Case of the Green Kitchen

Read each little story. Think for a moment. Write your answer in one or two words. The key waits on the next page, so no peeking.

CARD 01

COOKING WITH AIR

EVIDENCE

Photosynthesis: Rohan waters his pot and sets it under the bright sky. The plant drinks from below and catches bright rays from above. It also pulls a certain gas straight from the breeze around it.

Q. Which gas taken from the air do plants use along with sunlight and water?

YOUR ANSWER _____

CARD 02

THE GREEN TRAP

EVIDENCE

Photosynthesis: Meera looks closely at a fresh hedge in the school yard. A special substance gives the flat blades their bright colour. This substance catches the bright warmth beating down on them.

Q. What is the green pigment in leaves that traps sunlight energy called?

YOUR ANSWER _____

CARD 03

SWEET OUTPUT

EVIDENCE

Photosynthesis: A tiny shoot works quietly all morning under the open sky. By noon, its green cells have finished cooking a sweet simple meal. The shoot stores this freshly made sweet stuff to grow.

Q. What is the sugar made by plants called?

YOUR ANSWER _____

CARD 04

A BREATH FOR THE WORLD

EVIDENCE

Photosynthesis: Kabir sits under a leafy tree during a warm afternoon. While the foliage works in the sun, it sends a fresh gas straight out into the breeze. All passing animals take in this fresh air.

Q. Which gas is given out into the air by the plant?

YOUR ANSWER _____

CARD 05

UNDER THE MUD

EVIDENCE

Parts That Help: Priya pours a mug of moisture into the dry brown earth of her flowerpot. Down below the surface, hidden underground parts drink up every drop. Soon the moist ground becomes dry again.

Q. Which parts take in water from the soil?

YOUR ANSWER _____

CARD 06

MICROSCOPIC OPENINGS

EVIDENCE

Parts That Help: Arjun turns a broad blade upside down to examine its skin. Invisible tiny openings sit across its surface. The breeze slips into the blade right through these minute gaps.

Q. What are the tiny pores on a leaf called?

YOUR ANSWER _____

CARD 07

PIPELINES ON A SHEET

EVIDENCE

Parts That Help: Riya holds an old fallen leaf against the window. Fine thin lines spread like little roads across the blade. They move liquid up to the top and carry meals back out to the stem.

Q. What are these structures that carry water and food called?

YOUR ANSWER _____

CARD 08

KITCHEN IN THE SUN

EVIDENCE

A potted bush stands on Asha's balcony. Nobody feeds it soup or bread, yet it grows big and healthy. The green bush makes its own meals.

Q. What is this process by which green plants make their own food called?

YOUR ANSWER _____

Answer Key · Sheet 1

For the parent. Keep this page folded away; open it only after every answer is written down.

01
COOKING WITH AIR
Carbon dioxide
Plants use sunlight, water and carbon dioxide from the air to make food.
Sheet says: p. 1: They use sunlight, water and carbon dioxide from the air.

02
THE GREEN TRAP
Chlorophyll
Chlorophyll is the green pigment in leaves that traps the energy of sunlight.
Sheet says: p. 1: Chlorophyll, the green pigment in leaves, traps the energy of sunlight.

03
SWEET OUTPUT
Glucose
The food green plants make is a sugar called glucose.
Sheet says: p. 1: The food they make is a sugar called glucose.

04
A BREATH FOR THE WORLD
Oxygen
Oxygen is given out into the air during photosynthesis.
Sheet says: p. 1: Oxygen is given out into the air.

05
UNDER THE MUD
Roots
Roots take in water from the soil.
Sheet says: p. 1: Roots take in water from the soil.

06
MICROSCOPIC OPENINGS
Stomata
Carbon dioxide enters the leaf through tiny pores called stomata.
Sheet says: p. 1: Carbon dioxide enters the leaf through tiny pores called stomata.

07
PIPELINES ON A SHEET
Veins
Veins carry water to the leaves and food to the rest of the plant.
Sheet says: p. 1: Veins carry water to the leaves and food to the rest of the plant.

08
KITCHEN IN THE SUN
Photosynthesis
Green plants make their own food by photosynthesis.
Sheet says: p. 1: Green plants make their own food by photosynthesis.

YOUR SCORE

all right	**CASE CLOSED**	a true detective
most right	**SHARP EYE**	reread a card or two
a few right	**KEEP GOING**	read the little stories again, slowly

CASE CLOSED